Cultural Heritage Visualization System — Installation Guide

(Руководство по установке)

1. Windows Installation Guide

Prerequisites (Требования):

- Python 3.8+ (Ensure "Add Python to PATH" is checked).
- MySQL 8.0+ Community Server.

Step 1: Database Setup

1. Open MySQL Workbench.
2. Execute the SQL script located at `database/init.sql` to create the database and table.
3. Execute the SQL script located at `database/seed_data.sql` to insert the heritage data.

Step 2: Backend Setup

1. Open Command Prompt (cmd) and navigate to the project folder:

   ```
   cmd

   cd  path\to\heritage_web
   ```
2. Install dependencies:

   ```
   cmd

   pip  install  -r  requirements.txt
   ```
3. Create a `.env` file in the root directory and configure your database credentials:

   ```
   env

   DB_HOST=localhost
   DB_USER=root
   DB_PASSWORD=your_mysql_password
   DB_NAME=heritage_db
   ```
4. Start the Flask server:

   ```
   cmd

   python  app.py
   ```

Step 3: Frontend Access
Double-click `templates/index.html` to open it in your browser, or use VS Code's "Live Server" extension.

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2. Linux (Fedora) Installation Guide

Prerequisites (Требования):
Open the terminal and run the following commands to install necessary packages:

```
bash
sudo dnf update -y
sudo dnf install python3 python3-pip mariadb-server -y
```

Step 1: Database Setup

1. Start and enable MariaDB (MySQL fork):

   ```
   bash

   sudo systemctl start mariadb
   sudo systemctl enable mariadb
   ```
2. Secure the database installation:

   ```
   bash

   sudo mysql_secure_installation
   ```
3. Log in to MariaDB and execute the SQL scripts located in `database/init.sql` and `database/seed_data.sql`.

Step 2: Backend Setup

1. Navigate to the project directory:

   ```
   bash

   cd heritage_web
   ```
2. Install dependencies:

   ```
   bash

   pip3 install -r requirements.txt
   ```
3. Configure the `.env` file with your MariaDB password.

4. **IMPORTANT: Configure Firewall for Fedora**
   To allow access to port 5000, run:

   bash

   ```
   sudo firewall-cmd --zone=public --add-port=5000/tcp --permanent
   sudo firewall-cmd --reload
   ```
5. Start the server:

   bash

   ```
   python3 app.py
   ```

Step 3: Frontend Access
Open the templates/index.html file in your browser. Ensure the backend is running at http://127.0.0.1:5000.

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3. User Guide (Руководство пользователя)

Interface Overview:

- **Language Switcher**: Located at the top-right corner (EN / RU / 中文).

  Click to switch languages instantly.
- **Filter Dropdowns**: Select historical dynasties and heritage types.
- **Search Box**: Input keywords to find heritage by name.
- **Action Buttons**: Click "Search" to apply filters, or "Reset" to clear all filters.

How to Use:

1. **Searching**: Select a dynasty or type, or type a keyword, then click "Search". The map refreshes to show matching results.
2. **Viewing Details**: Hover over or click on any blue marker on the map. A pop-up window will display the heritage's details in the selected language.
3. **Resetting**: Click "Reset" to restore the full-data view.